P15 Urn has 1 white and 1 black

Each time, choose a ball from urn and put back two balls with the same colour into the urn.

Aim: Find prob. that i of n balls are white.

When $n=2$, ~~prob~~ $P(1$ ~~are~~ is white$) = 1$

~~When $n \geq 3$~~ Let Q_n : $P(i$ are white when we have n balls$) = \frac{1}{n-1}$ $\forall i = 1, \dots, n-1$

$\uparrow$ statement

When $n=3$, $P(1$ is white$) = \frac{1}{2}$ (prob. to get black ball)

$P(2$ ~~is~~ are white$) = \frac{1}{2}$ (prob. to get white ball)

$\therefore Q_3$ is true (base case)

Assume that Q_k is true for some $k \geq 3$.

That is, $P(i \text{ are white and } k \text{ balls}) = \frac{1}{k-1} \quad \forall i = 1, \dots, k-1.$

Consider the urn with $k+1$ balls.
Let i be a number between 1 and k.

$\therefore P(i \text{ are white and } k+1 \text{ balls})$

$= P(i-1 \text{ are white and } k \text{ balls}) P(\text{Get white ball from } k \text{ balls})$ (while $i-1$ white)

$\quad + P(i \text{ are white and } k \text{ balls}) P(\text{Get black ball from } k \text{ balls})$ ↳ i are white

$$= \frac{1}{k-1} \cdot \frac{i-1}{k} + \frac{1}{k-1} \cdot \frac{k-i}{k}$$

$$= \frac{1}{(k-1)k} \cdot (i-1+k-i)$$

$$= \frac{1}{k} \quad \Rightarrow Q_{k+1} \text{ is true}$$

$\therefore P(i \text{ are white}) = \frac{1}{n-1} \quad \forall i = 1, \dots, n-1$
and n balls

by induction.

#Note Actually the events are easier to describe once we learn random variable.

Let X_n be a random variable representing # white given n balls.

Let E_i be the event of favourable outcomes.

In fact, $E_i = \{X_n = i\}$

Note that $E_i = (E_i \cap \{X_{n-1} = i\}) \cup (E_i \cap \{X_{n-1} = i-1\})$

Here I start with $\Omega = \bigcup_{k=1}^{n-2} \{X_{n-1} = k\}$ and I see that $E_i \cap \{X_{n-1} = k\} = \phi$ when $k \neq i, i-1$.

$\therefore P(E_i) = \cancel{P(\{X_n = i\})}$

$$= P(E_i \cap \{X_{n-1} = i\}) + P(E_i \cap \{X_{n-1} = i-1\})$$

$$= P(E_i \mid X_{n-1} = i)\, P(X_{n-1} = i) + P(E_i \mid X_{n-1} = i-1)\, P(X_{n-1} = i-1)$$

$P(E_i \mid X_{n-1} = k)$ is the prob. that we get i balls when we have $n-1$ balls and k white balls.

2nd Approach

$P(1 \text{ white given } n \text{ balls}) = $ ~~P(Black given 2 balls)~~

$$= P(\text{Black} \mid X_2 = 1)\, P(\text{Black} \mid X_3 = 1) \cdots P(\text{Black} \mid X_{n-1} = 1)$$

$$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n-2}{n-1} = \frac{1}{n-1}$$

Now let i be any number between 1 and $n-1$

$P(i \text{ white given } n \text{ balls})$

$$= \sum P(T_1 \mid X_2 = k_1)\, P(T_2 \mid X_3 = k_2) \cdots P(T_{n-2} \mid X_{n-1} = k_{n-2})$$

$$= \sum \prod_{j=1}^{n-2} P(T_j \mid X_{j+1} = k_j)$$

where the summation goes over sequences $(T_1, \ldots, T_{n-2})$ of white and black containing exactly $i-1$ white and $(k_1, \ldots, k_{n-2})$ is determined by $(T_1, \ldots, T_{n-2})$, that is $k_1 = 1$ and

$$k_j = \begin{cases} k_{j-1} + 1 & \text{if } T_{j-1} = \text{white}, \\ k_{j-1} & \text{if } T_{j-1} = \text{black}. \end{cases} \quad \text{for } j = 2, \ldots, n-2$$

Since there are $i-1$ white in $(T_1, \ldots, T_{n-2})$, we see that

$$\prod_{j=1}^{n-2} P(T_j \mid X_{j+1} = k_j)$$

$$= P(W \mid X_{j_1'+1} = k_{j_1'}) \cdots P(W \mid X_{j_{i-1}'+1} = k_{j_{i-1}'})$$

$$P(B \mid X_{j_1''+1} = k_{j_1''}) \cdots P(B \mid X_{j_{n-i-1}''+1} = k_{j_{n-i-1}''})$$

where $j_1' < j_2' < \cdots < j_{i-1}'$

$j_1'' < j_2'' < \cdots < j_{n-i-1}''$ and $\{1, 2, \ldots, n-2\} = \{j_1', \ldots, j_{i-1}'\} \cup \{j_1'', \ldots, j_{n-i-1}''\}$

Note that $k_{j_1'} = 1$ and all $k_{j_\ell'}$ are distinct

(If $k_{j_\ell'} = k_{j_m'}$ for $\ell < m$, then $k_{j_\ell'} = k_{j_{\ell+1}'} \Rightarrow T_{j_\ell'}$ is black)

~~Since~~ $\therefore\ k_{j_1'} = 1,\ k_{j_2'} = 2,\ \ldots,\ k_{j_{i-1}'} = i-1$

$$\therefore\ P(W \mid X_{j_1'+1} = k_{j_1'}) \cdots P(W \mid X_{j_{i-1}'+1} = k_{j_{i-1}'})$$

$$= \frac{(i-1)!}{(j_1'+1)(j_2'+1)\cdots(j_{i-1}'+1)}$$

$$\boxed{\text{Note: } P(W \mid X_n = k) = \frac{k}{n}}$$

Similarly $P(B \mid \cdots) \cdots P(B \mid \cdots)$

$$= \frac{(n-i-1)!}{(j_1''+1)\cdots(j_{n-i-1}''+1)}$$

$$\therefore\ P(W \mid \cdots) \cdots P(W \mid \cdots) P(B \mid \cdots) \cdots P(B \mid \cdots)$$

$$= \frac{(i-1)!\,(n-i-1)!}{(n-1)!}$$

$\hookrightarrow$ because $\{2, 3, \ldots, n-1\} = \{j_\ell' + 1\} \cup \{j_\ell'' + 1\}$

$$\therefore \prod_{j=1}^{n-2} P(T_j \mid X_{j+1} = k_j) = \frac{(i-1)!\,(n-i-1)!}{(n-1)!}$$

for all such sequences $(T_1, \ldots, T_{n-2})$.

Since there are $\binom{n-2}{i-1}$ sequences containing exactly $i-1$ white, we get

$$\begin{aligned}
&P(i \text{ white given } n \text{ balls}) \\
&= \sum \prod_{j=1}^{n-2} P(T_j \mid X_{j+1} = k_j) \\
&= \binom{n-2}{i-1} \frac{(i-1)!\,(n-i-1)!}{(n-1)!} \\
&\quad \vdots \\
&= \frac{1}{n-1}
\end{aligned}$$